

Tree-level differential cross section for Higgs-boson elastic scattering

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Fundamental to the Standard Model of Particle Physics and QFT, the extremely rare scattering process that involves two spin-0 scalar Higgs Bosons could potentially provide new insights to values such as the Higgs self-coupling; thus here we carry out a detailed calculation at the tree-level for this process. In particular, we compute the tree-level differential cross section for elastic Higgs-boson scattering, $hh \rightarrow hh$, in the Standard Model. Starting from the Higgs self-interactions after electroweak symmetry breaking, we derive the contact, s -, t -, and u -channel contributions to the invariant amplitude and express the result in terms of the Mandelstam variables and the center-of-mass scattering angle. We then obtain the labeled and identical-particle differential cross-section conventions, emphasizing the final-state symmetry factor for two indistinguishable Higgs bosons. The resulting amplitude agrees with standard literature expressions for Standard Model Higgs-Higgs elastic scattering.

I. INTRODUCTION

Higgs-boson pair dynamics probe the self-interaction structure of the Standard Model scalar sector. Although direct Higgs-boson scattering is not an elementary collider observable in isolation, the process $hh \rightarrow hh$ provides a clean field-theoretic setting in which the cubic and quartic Higgs self-couplings enter at tree level. It is therefore useful as a benchmark calculation for normalization conventions, identical-particle phase space, and the relation between Feynman rules and differential cross sections. We use standard QFT normalizations for amplitudes, propagators, and phase space [1, 2]. Related Standard Model amplitudes and Higgs-Higgs interactions have been discussed in Refs. [3–5].

We consider the elastic process

$$h(p_1) + h(p_2) \longrightarrow h(p_3) + h(p_4), \quad (1)$$

where all external particles are physical Higgs bosons with mass m_h . The calculation is performed in the center-of-mass frame. For a generic two-body process, the standard center-of-mass differential cross section is

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}} = \frac{1}{64\pi^2 s} \frac{|\vec{p}_f|}{|\vec{p}_i|} |\mathcal{M}|^2, \quad s = E_{\text{CM}}^2. \quad (2)$$

The quantity $|\mathcal{M}|^2$ in Eq. (2) is the absolute square of the invariant scattering amplitude. To see precisely where it enters, begin with the relativistic normalization of a one-particle state,

$$\langle \vec{p}' | \vec{p} \rangle = 2E_{\vec{p}} (2\pi)^3 \delta^{(3)}(\vec{p}' - \vec{p}).$$

With this convention, the interacting part of the S -matrix element for a $2 \rightarrow 2$ process is defined by

$$\langle p_3, p_4 | S - 1 | p_1, p_2 \rangle = i(2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4) \mathcal{M}.$$

In perturbative QFT this matrix element is generated by

the Dyson expansion

$$\begin{aligned} S &= T \exp \left(i \int d^4x \mathcal{L}_{\text{int}}(x) \right) \\ &= 1 + \sum_{n=1}^{\infty} \frac{i^n}{n!} \int d^4x_1 \cdots d^4x_n \\ &\quad \times T \{ \mathcal{L}_{\text{int}}(x_1) \cdots \mathcal{L}_{\text{int}}(x_n) \}. \end{aligned}$$

Wick's theorem converts the time-ordered products into contractions, and the LSZ reduction formula removes the external propagators and places the external momenta on shell. The connected, amputated contribution that remains is $i\mathcal{M}$; applying the Feynman rules is the diagrammatic way of performing these Dyson, Wick, and LSZ steps. Thus $\mathcal{M}$ is the coefficient left after the universal four-momentum conserving delta function has been removed. It contains the dynamics: coupling constants, propagators, vertex factors, and the coherent sum of all Feynman diagrams contributing at the chosen perturbative order.

A probability is obtained from the absolute square of the S -matrix element. Consequently, the dynamical factor is

$$|\mathcal{M}|^2 \equiv \mathcal{M} \mathcal{M}^*.$$

The square of the delta function is treated by placing the process in a large spacetime box; one factor of the delta function remains to enforce four-momentum conservation, while the other produces the total observation time T and normalization volume V . Symbolically,

$$\left[(2\pi)^4 \delta^{(4)}(P_f - P_i) \right]^2 = (2\pi)^4 \delta^{(4)}(P_f - P_i) VT.$$

Dividing the transition probability by that time and by the incident particle flux gives the invariant differential cross-section formula

$$d\sigma = \frac{|\mathcal{M}|^2}{4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}} d\Phi_2,$$

where the Lorentz-invariant two-body phase-space element is

$$d\Phi_2 = (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4) \times \frac{d^3\vec{p}_3}{(2\pi)^3 2E_3} \frac{d^3\vec{p}_4}{(2\pi)^3 2E_4}.$$

The bar indicates a sum over unobserved final spin and internal quantum numbers and an average over the corresponding initial quantum numbers:

$$\overline{|\mathcal{M}|^2} = \frac{1}{N_{\text{initial}}} \sum_{\substack{\text{initial states} \\ \text{final states}}} |\mathcal{M}|^2.$$

For the present process every external Higgs boson is a neutral spin-0 particle, so there are no spin or color labels to sum or average. Therefore $\overline{|\mathcal{M}|^2} = |\mathcal{M}|^2$.

It is essential that amplitudes are added before taking the square. If $\mathcal{M} = \sum_a \mathcal{M}_a$, where a labels the contact, s -, t -, and u -channel diagrams, then

$$|\mathcal{M}|^2 = \sum_a |\mathcal{M}_a|^2 + 2 \sum_{a < b} \text{Re}(\mathcal{M}_a \mathcal{M}_b^*).$$

The second term contains quantum interference and is why one cannot calculate the cross section by merely adding the individual diagram probabilities. Even when the tree-level amplitude is real in the physical region, the notation $|\mathcal{M}|^2$ remains appropriate because loop corrections, finite widths, and pole prescriptions can make amplitudes complex.

In the center-of-mass frame, the invariant incident-flux factor and the two-body phase space reduce as follows:

$$d\Phi_2 = \frac{1}{(2\pi)^2} \frac{p_f^2 dp_f d\Omega}{4E_3 E_4} \delta(\sqrt{s} - E_3 - E_4) = \frac{1}{16\pi^2} \frac{|\vec{p}_f|}{\sqrt{s}} d\Omega,$$

where the second line follows by using the energy delta function to perform the radial p_f integral. The flux factor becomes

$$\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2} = |\vec{p}_i| \sqrt{s}.$$

Substitution into the invariant formula gives

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\vec{p}_f|}{|\vec{p}_i|} |\mathcal{M}|^2,$$

which is Eq. (2) for the spinless Higgs process. In natural units, a cross section has mass dimension -2 . Since the prefactor $1/s$ already has that dimension, the four-dimensional invariant amplitude $\mathcal{M}$ for this renormalizable $2 \rightarrow 2$ process is dimensionless.

For equal incoming and outgoing masses,

$$m_1 = m_2 = m_3 = m_4 = m_h, \quad |\vec{p}_f| = |\vec{p}_i| \equiv p, \quad (3)$$

so Eq. (2) becomes, for labeled final-state momenta,

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}}^{\text{labeled}} = \frac{1}{64\pi^2 s} |\mathcal{M}|^2. \quad (4)$$

The word “labeled” means that p_3 and p_4 are treated as distinguishable labels in the phase-space parametrization. Physically, the two final Higgs bosons are identical. If one integrates over the full solid angle without distinguishing which final Higgs is assigned label 3, the usual final-state symmetry factor $1/2!$ must be included. Thus the corresponding unlabelled full-solid-angle convention is

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}}^{\text{identical}} = \frac{1}{2} \frac{1}{64\pi^2 s} |\mathcal{M}|^2 = \frac{1}{128\pi^2 s} |\mathcal{M}|^2. \quad (5)$$

Equivalently, one may keep Eq. (4) and integrate over one representative half of the final-state angular phase space. This is the convention underlying the existing main results used here.

II. STANDARD MODEL HIGGS SELF-INTERACTIONS

The full Standard Model Lagrangian can be organized schematically as

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Yukawa}} + \mathcal{L}_{\text{Higgs}},$$

where the part needed to derive Higgs self-interactions is

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi).$$

Here Φ is a complex scalar doublet of $SU(2)_L$ with weak hypercharge $Y = 1/2$, and

$$D_\mu = \partial_\mu - i \frac{g}{2} \tau^a W_\mu^a - i g' Y B_\mu$$

is the electroweak covariant derivative. The kinetic term supplies the canonically normalized Higgs kinetic energy and its gauge interactions. The non-derivative Higgs self-interactions come entirely from $-V(\Phi)$.

Renormalizability permits operators of mass dimension at most four. Gauge invariance requires the potential to depend on the singlet $\Phi^\dagger \Phi$. Up to an irrelevant additive constant, the most general renormalizable choice is therefore the quadratic-plus-quartic potential

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2. \quad (6)$$

For $\mu^2 > 0$ and $\lambda > 0$, the negative quadratic term makes the symmetric point $\Phi = 0$ unstable, while the positive quartic term stabilizes the potential. Before choosing a gauge, fluctuations about the symmetry-breaking vacuum may be written as

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{2} G^+(x) \\ v + h(x) + i G^0(x) \end{pmatrix},$$

where $G^\pm$ and G^0 are the three Goldstone fields. In unitary gauge these fields are removed from the scalar doublet and become the longitudinal degrees of freedom of $W^\pm$ and Z . The scalar doublet then becomes

$$\Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v+h \end{pmatrix}, \quad (7)$$

where $v \simeq 246 \text{ GeV}$ is the Higgs vacuum expectation value and h is the physical Higgs field. The kinetic term then contains

$$(D_\mu \Phi)^\dagger (D^\mu \Phi) \supset \frac{1}{2} (\partial_\mu h)(\partial^\mu h),$$

which confirms that h has canonical normalization.

We now derive the potential for h explicitly. Equation (7) gives

$$\Phi^\dagger \Phi = \frac{(v+h)^2}{2},$$

and substitution into Eq. (6) yields

$$V(h) = -\frac{\mu^2}{2}(v+h)^2 + \frac{\lambda}{4}(v+h)^4.$$

Expanding both powers without yet imposing the vacuum condition gives

$$V(h) = -\frac{\mu^2 v^2}{2} + \frac{\lambda v^4}{4} + (-\mu^2 v + \lambda v^3)h + \left(-\frac{\mu^2}{2} + \frac{3\lambda v^2}{2}\right)h^2 + \lambda v h^3 + \frac{\lambda}{4}h^4.$$

The vacuum expectation value is defined by stationarity at $h=0$:

$$\left. \frac{dV}{dh} \right|_{h=0} = -\mu^2 v + \lambda v^3 = 0.$$

For the broken vacuum $v \neq 0$, this condition implies

$$\mu^2 = \lambda v^2.$$

It removes the term linear in h , as required when expanding about a true vacuum. The physical Higgs mass is read from the coefficient of the canonical mass term $-m_h^2 h^2/2$ in the Lagrangian, equivalently $+m_h^2 h^2/2$ in the potential:

$$m_h^2 = \left. \frac{d^2 V}{dh^2} \right|_{h=0} = -\mu^2 + 3\lambda v^2 = 2\lambda v^2.$$

Using $\lambda = m_h^2/(2v^2)$ in the quadratic, cubic, and quartic terms gives

$$\begin{aligned} \lambda v^2 h^2 &= \frac{1}{2} m_h^2 h^2, \\ \lambda v h^3 &= \frac{m_h^2}{2v} h^3, \\ \frac{\lambda}{4} h^4 &= \frac{m_h^2}{8v^2} h^4. \end{aligned}$$

Therefore, expanding around the minimum gives

$$V(h) = \text{constant} + \frac{1}{2} m_h^2 h^2 + \frac{m_h^2}{2v} h^3 + \frac{m_h^2}{8v^2} h^4, \quad (8)$$

with

$$m_h^2 = 2\lambda v^2. \quad (9)$$

Because $\mathcal{L}_{\text{int}} = -V_{\text{int}}$,

$$\mathcal{L}_{\text{int}} = -\frac{m_h^2}{2v} h^3 - \frac{m_h^2}{8v^2} h^4. \quad (10)$$

Writing the interaction Lagrangian in the standard Feynman-rule normalization,

$$\mathcal{L}_{\text{int}} = -\frac{g_3}{3!} h^3 - \frac{g_4}{4!} h^4, \quad (11)$$

one obtains

$$g_3 = \frac{3m_h^2}{v}, \quad g_4 = \frac{3m_h^2}{v^2}. \quad (12)$$

The relevant Feynman rules are therefore

$$hhh : \quad -ig_3 = -i\frac{3m_h^2}{v}, \quad hhhh : \quad -ig_4 = -i\frac{3m_h^2}{v^2}. \quad (13)$$

III. TREE-LEVEL AMPLITUDE

At tree level the amplitude receives four contributions: one contact diagram from the quartic vertex and one Higgs-exchange diagram in each of the s , t , and u channels. The Mandelstam variables are

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2, \quad (14)$$

and for equal external masses they obey

$$s + t + u = 4m_h^2. \quad (15)$$

The contact contribution is

$$i\mathcal{M}_{\text{contact}} = -ig_4, \quad \mathcal{M}_{\text{contact}} = -g_4. \quad (16)$$

For the s -channel diagram, the two cubic vertices and the internal Higgs propagator give

$$i\mathcal{M}_s = (-ig_3) \frac{i}{s - m_h^2 + i\epsilon} (-ig_3). \quad (17)$$

Since $(-ig_3)^2 = -g_3^2$,

$$i\mathcal{M}_s = -i \frac{g_3^2}{s - m_h^2 + i\epsilon}, \quad \mathcal{M}_s = -\frac{g_3^2}{s - m_h^2 + i\epsilon}. \quad (18)$$

Similarly,

$$\mathcal{M}_t = -\frac{g_3^2}{t - m_h^2 + i\epsilon}, \quad \mathcal{M}_u = -\frac{g_3^2}{u - m_h^2 + i\epsilon}. \quad (19)$$

The invariant matrix element is the sum of the four tree diagrams,

$$\mathcal{M} = \mathcal{M}_{\text{contact}} + \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u. \quad (20)$$

Substituting the four contributions gives

$$\mathcal{M} = -g_4 - \frac{g_3^2}{s - m_h^2 + i\epsilon} - \frac{g_3^2}{t - m_h^2 + i\epsilon} - \frac{g_3^2}{u - m_h^2 + i\epsilon} \quad (21)$$

$$= -g_4 - g_3^2 \left(\frac{1}{s - m_h^2 + i\epsilon} + \frac{1}{t - m_h^2 + i\epsilon} + \frac{1}{u - m_h^2 + i\epsilon} \right). \quad (22)$$

The full tree-level amplitude is therefore

$$\mathcal{M}(s, t, u) = -g_4 - g_3^2 \left(\frac{1}{s - m_h^2 + i\epsilon} + \frac{1}{t - m_h^2 + i\epsilon} + \frac{1}{u - m_h^2 + i\epsilon} \right). \quad (23)$$

Substituting Eq. (12) gives

$$\mathcal{M}_{\text{SM}} = -\frac{3m_h^2}{v^2} - \left(\frac{3m_h^2}{v} \right)^2 \left(\frac{1}{s - m_h^2 + i\epsilon} + \frac{1}{t - m_h^2 + i\epsilon} + \frac{1}{u - m_h^2 + i\epsilon} \right) \quad (24)$$

$$= -\frac{3m_h^2}{v^2} - \frac{9m_h^4}{v^2} \left(\frac{1}{s - m_h^2 + i\epsilon} + \frac{1}{t - m_h^2 + i\epsilon} + \frac{1}{u - m_h^2 + i\epsilon} \right). \quad (25)$$

Factoring out $-3m_h^2/v^2$ gives

$$\boxed{\mathcal{M}_{\text{SM}}(s, t, u) = -\frac{3m_h^2}{v^2} \left[1 + 3m_h^2 \left(\frac{1}{s - m_h^2 + i\epsilon} + \frac{1}{t - m_h^2 + i\epsilon} + \frac{1}{u - m_h^2 + i\epsilon} \right) \right]}. \quad (26)$$

For the physical elastic region $s \geq 4m_h^2$, none of the denominators reaches a single-Higgs pole for real scattering angles, so the $i\epsilon$ prescription can usually be left implicit. Equation (26) matches the standard tree-level Standard Model result quoted in Refs. [3, 4].

IV. CENTER-OF-MASS KINEMATICS AND ANGULAR DEPENDENCE

In the center-of-mass frame,

$$p_1 = (E, 0, 0, p), \quad p_2 = (E, 0, 0, -p), \quad (27)$$

$$p_3 = (E, p \sin \theta, 0, p \cos \theta), \quad (28)$$

$$p_4 = (E, -p \sin \theta, 0, -p \cos \theta). \quad (29)$$

Each external Higgs is on shell,

$$E^2 = p^2 + m_h^2, \quad (30)$$

and

$$E_{\text{CM}} = 2E, \quad s = E_{\text{CM}}^2 = 4E^2. \quad (31)$$

It follows that

$$E = \frac{\sqrt{s}}{2}, \quad p^2 = \frac{s}{4} - m_h^2, \quad (32)$$

or, equivalently,

$$\beta = \sqrt{1 - \frac{4m_h^2}{s}}, \quad p = \frac{\sqrt{s}}{2}\beta. \quad (33)$$

Using the mostly-minus metric,

$$t = (p_1 - p_3)^2 = -(p \sin \theta)^2 - [p(1 - \cos \theta)]^2 \quad (34)$$

$$= -2p^2(1 - \cos \theta) = -\frac{s}{2}\beta^2(1 - \cos \theta), \quad (35)$$

and similarly

$$u = -2p^2(1 + \cos \theta) = -\frac{s}{2}\beta^2(1 + \cos \theta). \quad (36)$$

These expressions satisfy

$$s + t + u = s - 4p^2 = 4m_h^2. \quad (37)$$

Substituting Eqs. (35) and (36) into Eq. (26) proceeds denominator by denominator:

$$t - m_h^2 = -2p^2(1 - \cos \theta) - m_h^2, \quad (38)$$

$$u - m_h^2 = -2p^2(1 + \cos \theta) - m_h^2. \quad (39)$$

Hence the bracket in Eq. (26) becomes

$$1 + 3m_h^2 \left[\frac{1}{s - m_h^2} + \frac{1}{-2p^2(1 - \cos \theta) - m_h^2} + \frac{1}{-2p^2(1 + \cos \theta) - m_h^2} \right], \quad (40)$$

where the $i\epsilon$ terms have been suppressed away from the poles. Therefore

$$\mathcal{M}_{\text{SM}}(s, \theta) = -\frac{3m_h^2}{v^2} \left[1 + 3m_h^2 \left(\frac{1}{s - m_h^2} + \frac{1}{-2p^2(1 - \cos \theta) - m_h^2} + \frac{1}{-2p^2(1 + \cos \theta) - m_h^2} \right) \right]. \quad (41)$$

Using $2p^2 = (s - 4m_h^2)/2$, the two exchange denominators become

$$-2p^2(1 - \cos \theta) - m_h^2 = -\frac{s - 4m_h^2}{2}(1 - \cos \theta) - m_h^2, \quad (42)$$

$$-2p^2(1 + \cos \theta) - m_h^2 = -\frac{s - 4m_h^2}{2}(1 + \cos \theta) - m_h^2. \quad (43)$$

Equivalently,

$$\mathcal{M}_{\text{SM}}(s, \theta) = -\frac{3m_h^2}{v^2} \left[1 + 3m_h^2 \left(\frac{1}{s - m_h^2} + \frac{1}{-\frac{s - 4m_h^2}{2}(1 - \cos \theta) - m_h^2} + \frac{1}{-\frac{s - 4m_h^2}{2}(1 + \cos \theta) - m_h^2} \right) \right]. \quad (44)$$

This expression is invariant under $\cos \theta \rightarrow -\cos \theta$, as expected: the exchange of the two identical final Higgs bosons interchanges t and u .

Since $\mathcal{M}_{\text{SM}} = -(3m_h^2/v^2)B$, and the tree-level amplitude is real in the physical elastic region considered here,

V. DIFFERENTIAL CROSS SECTION

Combining Eq. (4) with the amplitude above gives the labeled center-of-mass differential cross section

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}}^{\text{labeled}} = \frac{1}{64\pi^2 s} |\mathcal{M}_{\text{SM}}(s, \theta)|^2. \quad (45)$$

Equivalently, in terms of the Mandelstam variables, define

$$B(s, t, u) = 1 + 3m_h^2 \left(\frac{1}{s - m_h^2} + \frac{1}{t - m_h^2} + \frac{1}{u - m_h^2} \right). \quad (46)$$

$$|\mathcal{M}_{\text{SM}}|^2 = \left(\frac{3m_h^2}{v^2} \right)^2 B(s, t, u)^2. \quad (47)$$

Substituting Eq. (47) into Eq. (4) gives

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}}^{\text{labeled}} = \frac{1}{64\pi^2 s} \left\{ \frac{3m_h^2}{v^2} \left[1 + 3m_h^2 \left(\frac{1}{s - m_h^2} + \frac{1}{t - m_h^2} + \frac{1}{u - m_h^2} \right) \right] \right\}^2. \quad (48)$$

with

$$t = -\frac{s - 4m_h^2}{2}(1 - \cos \theta), \quad u = -\frac{s - 4m_h^2}{2}(1 + \cos \theta). \quad (49)$$

For the identical-final-state full-phase-space convention, the phase-space symmetry factor multiplies the labeled

result:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}}^{\text{identical}} = \frac{1}{2!} \left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}}^{\text{labeled}} \quad (50)$$

$$= \frac{1}{2} \frac{1}{64\pi^2 s} |\mathcal{M}_{\text{SM}}(s, \theta)|^2. \quad (51)$$

Thus

$$\boxed{\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}}^{\text{identical}} = \frac{1}{128\pi^2 s} |\mathcal{M}_{\text{SM}}(s, \theta)|^2.} \quad (52)$$

The factor of 1/2 is purely kinematical: it prevents double counting of the same physical final state when the two outgoing Higgs bosons are indistinguishable.

VI. CONSISTENCY CHECKS

At threshold,

$$s = 4m_h^2, \quad p = 0, \quad t = u = 0. \quad (53)$$

The amplitude becomes

$$\mathcal{M}_{\text{thr}} = -\frac{3m_h^2}{v^2} \left[1 + 3m_h^2 \left(\frac{1}{4m_h^2 - m_h^2} + \frac{1}{0 - m_h^2} + \frac{1}{0 - m_h^2} \right) \right] \quad (54)$$

$$= -\frac{3m_h^2}{v^2} \left[1 + 3 \left(\frac{1}{3} - 1 - 1 \right) \right] \quad (55)$$

$$= \frac{12m_h^2}{v^2}. \quad (56)$$

Thus

$$\boxed{\mathcal{M}_{\text{thr}} = \frac{12m_h^2}{v^2}.} \quad (57)$$

This agrees with the standard tree-level threshold behavior of the Standard Model Higgs–Higgs scattering amplitude.

For $s \gg m_h^2$, away from the forward and backward regions, the exchange terms scale as

$$\frac{m_h^2}{s - m_h^2}, \quad \frac{m_h^2}{t - m_h^2}, \quad \frac{m_h^2}{u - m_h^2}. \quad (58)$$

The amplitude therefore approaches the quartic contact contribution,

$$\mathcal{M}_{\text{SM}}(s, \theta) \longrightarrow -\frac{3m_h^2}{v^2}, \quad (59)$$

and the labeled differential cross section falls as

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}}^{\text{labeled}} \longrightarrow \frac{1}{64\pi^2 s} \frac{9m_h^4}{v^4} \propto \frac{1}{s}. \quad (60)$$

VII. COMPARISON WITH LITERATURE

The amplitude in Eq. (26) coincides with the standard result for Standard Model Higgs–Higgs elastic scattering.

For example, Ref. [3] writes the tree-level amplitude as

$$\mathcal{A}_{\text{SM}}^{(0)} = -\frac{3M_H^2}{v^2} \left(1 + \frac{3M_H^2}{s - M_H^2} + \frac{3M_H^2}{t - M_H^2} + \frac{3M_H^2}{u - M_H^2} \right), \quad (61)$$

which is Eq. (26) after identifying $M_H = m_h$ and suppressing the $i\epsilon$ prescription away from poles. The same four tree diagrams and decomposition into contact, s -, t -, and u -channel terms are also discussed in Higgs–Higgs scattering studies such as Ref. [4].

VIII. CONCLUSION

We have derived the tree-level Standard Model amplitude and differential cross section for $hh \rightarrow hh$ in the center-of-mass frame. The calculation follows directly from the cubic and quartic Higgs self-interactions in the broken phase, leading to the compact amplitude in Eq. (26). The final cross-section formulas differ only by the conventional treatment of identical final-state phase space: Eq. (45) applies to labeled outgoing momenta, whereas Eq. (52) includes the full solid-angle identical-particle symmetry factor. The threshold and high-energy limits provide useful checks, and the amplitude agrees with published Standard Model expressions.

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